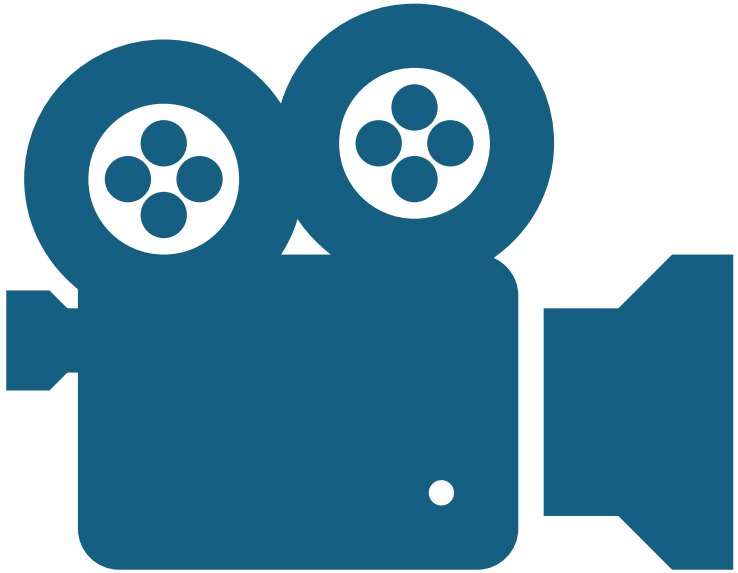


By: Alexa Varjian

AI in the Film World

AI History

- 1950s
  - Computers- extensive calculators
  - Turned to scientists when needing specific calculations
  - Alan Turing (British Mathematician)
    - Credited with the idea of AI
      - Did not have the tech needed to prove it

AI History

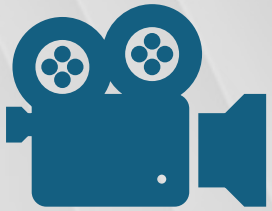
- Summer 1956
  - Professor John McCarthy
    - First coined the term “Artificial Intelligence”
    - Invited researchers to take part in summer workshop focused on possibility of AI
    - Group is credited for creating the AI field
      - 2 years after Alan Turing passed away

AI History

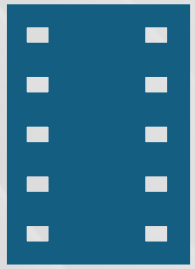
- ELIZA
  - Joseph Weizenbaum
  - Known as the first chatbot (1966)
  - Used to simulate therapy
    - Repurposed answers users gave and turned into questions that helped the conversation continue
- American Association of AI
  - 1980s

AI in Film

Pre-Production



On Set Production



Post-Production

Pre-Production

- Planning, scheduling and budgeting
- Repetitive and tedious tasks
  - Looking at data
- AI can help!
  - Many techniques used such as:
    - machine learning (ML)
    - computer vision
    - natural language processing (NLP)
  - Make process much easier
  - not as time consuming

Script Breakdown

- Dissecting a script
- Organizing everything into different categories based on elements
  - Actors
  - Props
  - Costumes
  - Locations
- Crucial for budget/time management
- Must look through script and tag everything
- *Very* tedious
- *NLP and ML can help!*

BREAKDOWN SHEET: # 1 | Page Count: 2 3/8s | Date:

Prod. Company:

Project Title: The Accomplice

Scene #: 1 | Scene Name: | INT/EXT: INT | D / N: NIGHT

Script Page: 1 | Location Name: Jake Pope's Apartment

Description: Jake Pope comes home to find the Thief

CAST	STUNTS	SET DRESSING
- The Thief -Detective Jake Pope		-Crime wall Candid photos, surv. shots, news paper clippings of Thief's crimes, etc. -Bedroom lamp -Bedroom and apt. decor -Table by front door -Light switch
	EXTRAS/SILENT	
SPECIAL EFFECTS	PROPS	VEHICLE / ANIMALS
	-tea kettle -tea mug -tea bag -Pope's gun -badge and keys	
WARDROBE	MAKEUP/HAIR	SOUND EFFECTS
-Gun holster for Jake		-tea kettle whistle -door rustling
SPECIAL EQUIPMENT	PRODUCTION NOTES	

studiobinder Create stripboards, shooting schedules, call sheets and more on studiobinder.com

AI: Script Breakdown

Natural language processing

- Helps AI understand, interpret, and produce human language
- ChatGPT
  - Uses NLP because when we ask a prompt, it can understand us and produce what we want them to

Machine learning

- Type of AI that allows computers to learn without explicitly being programmed on how to do something
- ChatGPT
  - Uses ML to know information without us telling them
  - “Break down this script into all its elements!”

Budget Estimation

- AI can look at different data to estimate this!
  - Historical finance data
  - Scripts
  - Actors
  - Locations
  - Film duration
- Uses this data to create rough film estimate
  - Using AI driven solution
- Compare budget of film to budgets of similar projects

Target Audience/Success or Fail

Uses analytics from past audience preferences to determine target audience

Helps producers and writers determine story development and who they're making this film for

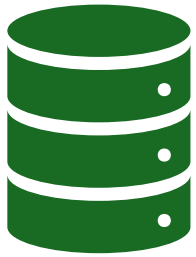
Can also predict the success (or lack thereof) of the film you're creating!

Using other scripts, box office history, and consumer data

Negatives

Ethical Considerations

Such as job displacement for many who can and will be replaced by AI



Transparency

Something AI generates could end up being false or biased

(Data quality or malfunction)

Can be problem for producers

AI in Screenwriting

- Three months for FIRST draft
- AI can do it in a couple minutes or even seconds!
- Example
  - Andy Herd creating *Friends* episode
- NLP/ML are used
  - Analyzes structure, style and semantics of existing scripts to find themes and it generates script based on what you tell it to write
- Can also show your work to AI and it can revamp it, identify plot holes, etc.
- Negative
  - Need to be very specific as to what you want out of it or else the result it produces will be awful
  - Limits creativity

AI in Screenwriting

- Andy Bobrow, tv writer and producer, did an experiment
  - Had AI help with scene
  - Told it the character names, location and that magic was involved
  - Said the result was lackluster
- Bobrow and Guy Goldstein (Writer Duet founder) say that AI lacks basic human quality needed to make stories good and worth reading/watching
- Goldstein says AI lacks an opinion and a “why”
  - Doesn’t understand why the story is important to tell
- AI should be used as an assistant to the writer, not THE writer

AI and On Set Production

- AI brings great help to film sets as well
  - Automating tasks like documenting, inventory checks, and data syncing
    - Lessens overhead costs and gives you opportunity to meet tight deadlines
  - Helps manage technical stuff so you and crew can focus your time on the quality of your film, the camera angles, etc.
  - Generates quick insights which can help prevent errors or missing certain shots
    - So, you won’t have to correct so many things in postproduction
  - AI analytics help enhance camera setups, lighting, and even your shot list
    - Once again allowing more time to focus on getting everything done that you need to and creating your best work!

On Set Production: Limitations

Limitations

Potential issues with data security

(Encryption is needed to avoid intellectual property theft or data breaches)

Potential impact on set production jobs

AI in Post-Production

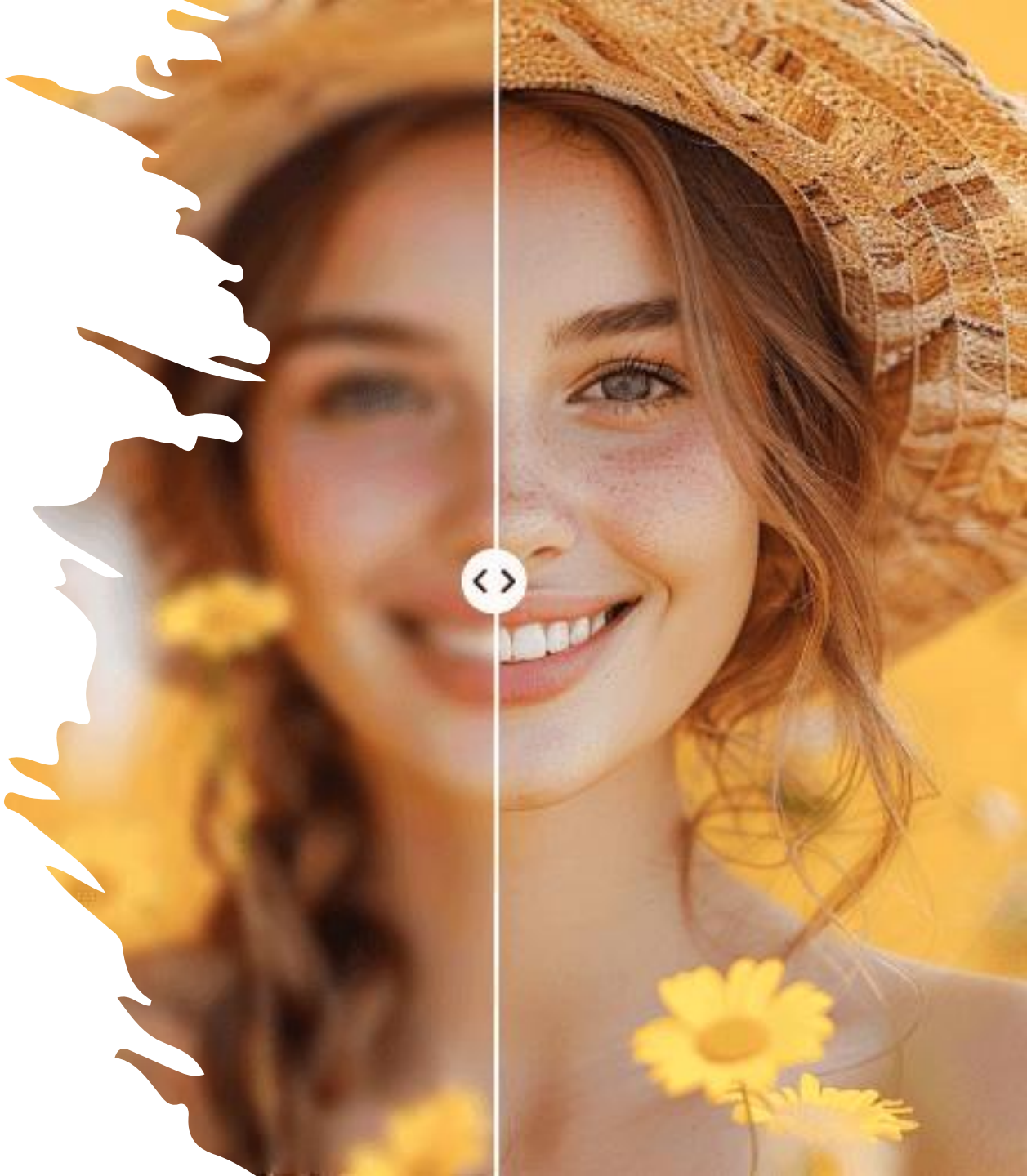
- What AI can assist with:
  - Add/Change dialogue
  - Before AI, we used ADR (Automated Dialogue Replacement)
    - Need to go back into the scene and re-articulate actor's mouth to say a new line and make it look natural
      - Super expensive/time consuming
    - Go back in and cut behind the shoulder of the character speaking or to a place where the dialogue can be said off screen
  - With AI, ADR has been changed (visual ADR)
      - Censorship editing
      - Dialogue replacement
      - Syncing same line replacement

Dubbing/Captioning/Translating

- Streaming services are now releasing content to audiences and using AI translation and sync tools
  - Helps keep the emotional nuance across all different languages
  - Example: Watch the Skies
    - Originally a Swedish film
    - TrueSync technology was used to make English version

Picture Improvement/Edit Acceleration

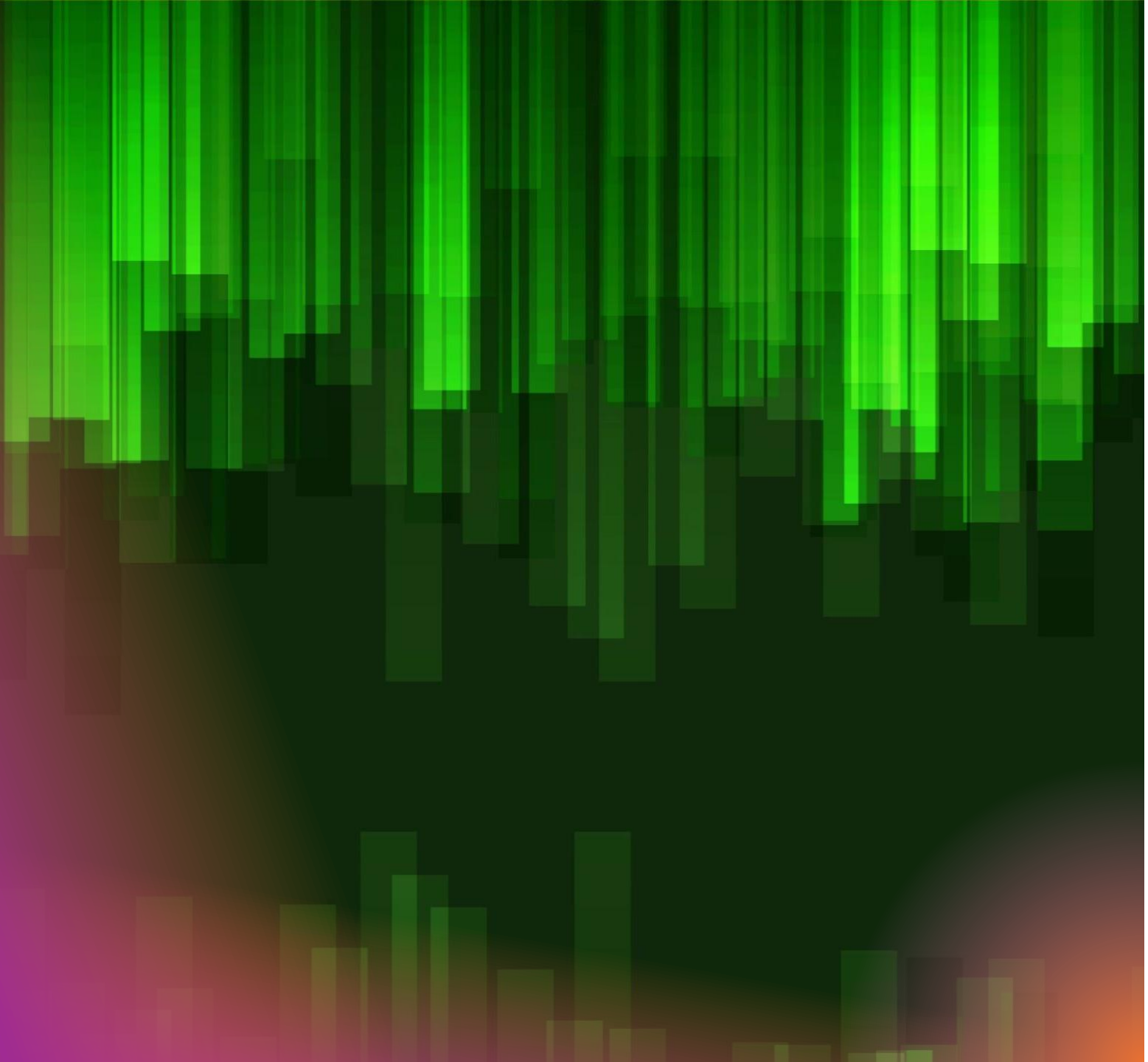
- AI is currently working to help restore frames that were once considered unusable
  - It can denoise (remove unwanted grain or discoloration), deblur, and upscale frames
- AI is becoming part of the nonlinear editing (NLE) process
  - Speeds everything up from dialogue transcription to scene search and audio cleanup



Object Removal/Continuity Fixes/AI Inpainting

Can help remove unwanted objects across moving footage

DeepEditor can handle motion blur and difficult lightings



Sound Design/Audio

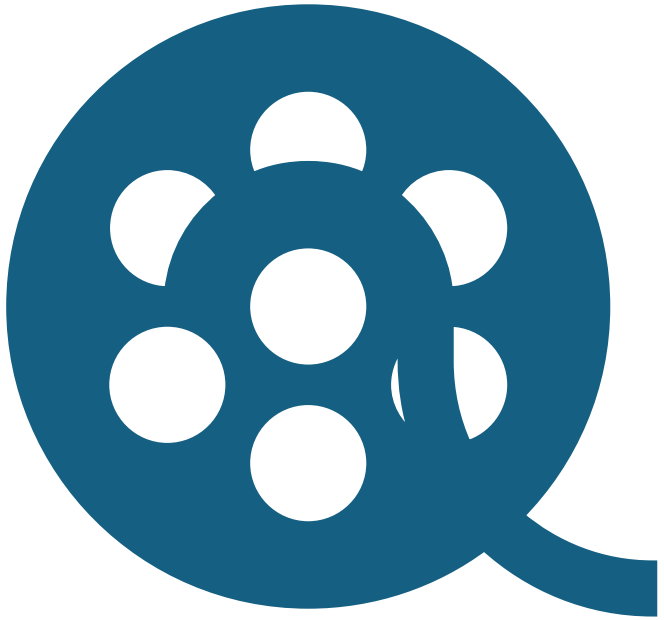
- Considered one of AI's most powerful uses in postproduction
- AI can:
  - Separate stems (split and extract audio files into separate files for each instrument)
  - Recreate missing ambience
  - Add sound effects into an already recorded environment
    - Makes added effects sound natural

Job Insecurity

- Study by Concept Art Association and the Animation Guild
  - Surveyed 300 people across entertainment industry
    - Study said the jobs most likely to be affected are:
      - Sound for film, tv, and video games
      - Developing 3-D assets
      - Foreign language dubbing
    - Study said the jobs least likely to be affected are:
      - Script writing
      - Performing music
    - 80% of early adopters of AI are using AI tools for postproduction purposes

The Flipside

- Not everyone believes AI will take over the film industry
- Tomas Elliott- Assistant professor of English at Northeastern University in London
    - Specializes in film history
    - “Cinema has navigated such huge transformations over its history, and in the long run these things tend to not have such widespread effects as the panic would suggest,” (Mello-Klein, 2024)
- Josh Rose- Filmmaking professor and cinematographer
    - “The amount of human time it takes to write a book, write a song, write a joke, create a choreography, write a symphony, or even take a photograph is, actually, one of the most important parts of the creation.” (Rose, 2025)



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Thank You!